

Week 3: The Avellaneda-Stoikov Model

Optimal market making via stochastic control – derived from scratch

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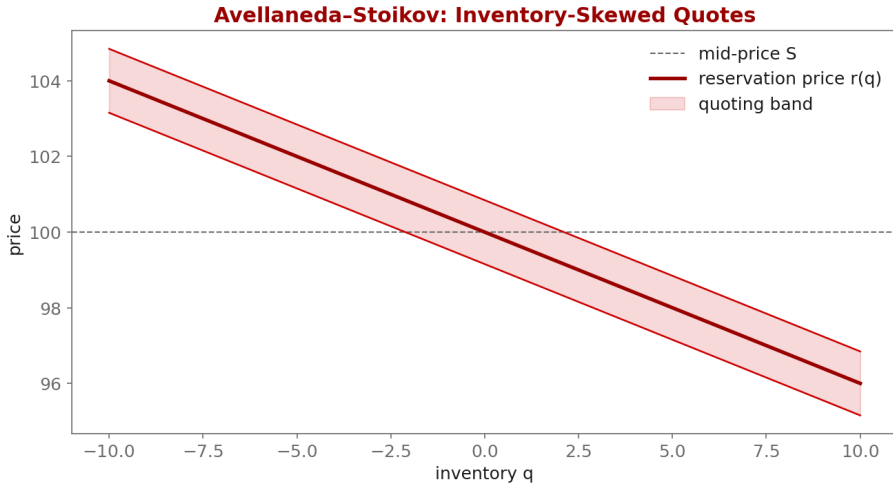
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1. Learning objectives
2. Lecture 1: Setup & Dynamics
3. Lecture 2: HJB, Ansatz, Closed Form
4. Illustration
5. Derivation & rationale
6. Getting the data from Coincall
7. Code: `week3_as_quotes.py`
8. Glossary of key terms
9. Reading
10. References

- ▶ Set up the Avellaneda-Stoikov control problem
- ▶ Follow the HJB derivation and understand every modeling choice
- ▶ Interpret the reservation price and the optimal half-spread

- ▶ Mid-price: $dS_t = \sigma dW_t$ (arithmetic Brownian motion)
- ▶ Fill intensity: $\lambda(\delta) = A \exp(-\kappa \delta)$
- ▶ Wealth from fills; inventory q jumps by ± 1 on a fill
- ▶ Exponential (CARA) utility of terminal wealth

- ▶ Dynamic programming $\rightarrow$ HJB partial differential equation
- ▶ Separation ansatz collapses HJB to ODEs in inventory
- ▶ Asymptotic (long-horizon) closed-form quotes
- ▶ Reservation price skews quotes against inventory



$$dS_t = \sigma dW_t$$

Why: Arithmetic (not geometric) Brownian motion is chosen because over a short market-making horizon price changes are small and roughly additive; it keeps the HJB linear in S and makes the cash dynamics exact.

$$\lambda(\delta) = A e^{-\kappa \delta}$$

Why: Fill intensity decays with quote distance δ . The exponential form is the simplest law consistent with 'further from mid $\Rightarrow$ fewer fills'; A sets the base rate, κ the decay. It also yields a closed-form optimum.

$$\max \mathbb{E} \left[-e^{-\gamma (X_T + q_T S_T)} \right]$$

Why: CARA (exponential) utility is used because constant absolute risk aversion makes the value function factor out wealth, so the optimal quotes do not depend on the cash level – exactly what we want for a market maker.

$$\partial_t V + \frac{1}{2} \sigma^2 \partial_{SS} V + \max_{\delta^b} \lambda(\delta^b) [V(q+1, x-S+\delta^b) - V] + \max_{\delta^a} \lambda(\delta^a) [V(q-1, x+S+\delta^a) - V] = 0$$

Why: The HJB equation: the diffusion term prices inventory risk; each max term is the dealer optimally choosing the bid/ask offset, trading fill probability against captured spread.

$$V(t, x, S, q) = -e^{-\gamma(x+qS)} \theta(t, q)$$

Why: This ansatz is not a guess from nowhere: under CARA + arithmetic dynamics the value function inherits a multiplicative structure across (x, qS) , so all the wealth/price dependence is exponential and only $\theta(t, q)$ remains to solve.

$$r(t, q) = S - q \gamma \sigma^2 (T - t)$$

Why: The reservation (indifference) price: it sits BELOW mid when you are long ($q > 0$) so you quote to sell down inventory. The skew scales with risk aversion γ , variance σ^2 , and time left – more risk $\Rightarrow$ more skew.

$$\delta^* = \frac{1}{\gamma} \ln\left(1 + \frac{\gamma}{\kappa}\right) + \frac{\gamma\sigma^2}{2}(T - t)$$

Why: The optimal half-spread = an order-flow term $\frac{1}{\gamma} \ln(1 + \gamma/\kappa)$ (how hard fills are to get) plus an inventory-risk term $\frac{\gamma\sigma^2}{2}(T - t)$. You widen when risk-averse, when volatile, and when far from the horizon.

- ▶ This week uses simulated dynamics (no live data needed).
- ▶ For realistic sigma, estimate it from Coincall perp klines:
 - ▶ GET /open/option/market/kline/history/v1/{name}?period=m1 (or futures klines)
- ▶ You will calibrate A , κ from real fills in Week 5.

Cross-check exact paths at <https://docs.coincall.com/>

```
import torch

def as_quotes(S, q, gamma, sigma, kappa, tau):
    """Asymptotic Avellaneda-Stoikov bid/ask. Vectorized over inventory q."""
    reservation = S - q * gamma * sigma**2 * tau
    half_spread = (1.0/gamma) * torch.log1p(gamma/kappa) \
        + 0.5 * gamma * sigma**2 * tau
    return reservation - half_spread, reservation + half_spread # bid, ask

S = torch.tensor(100.0); q = torch.arange(-10, 11).float()
bid, ask = as_quotes(S, q, gamma=0.1, sigma=2.0, kappa=1.5, tau=1.0)
print("inventory ->", q.tolist())
print("bid ->", bid.round(decimals=2).tolist())
```

Reservation price inventory-adjusted indifference price that quotes are centered on

Half-spread distance from the reservation price to each quote

HJB equation the PDE characterizing the optimal value function

Ansatz an assumed functional form that reduces the HJB to ODEs

CARA utility constant absolute risk aversion, $U(w) = -\exp(-\gamma w)$

Intensity $\lambda(\delta)$ Poisson fill rate as a function of quote distance δ

Risk aversion γ parameter setting inventory skew and spread width

- ▶ Avellaneda-Stoikov (2008); Cartea-Jaikumar-Penalva, Ch. 10

- ▶ Avellaneda, M., & Stoikov, S. (2008). High-frequency trading in a limit order book. *Quantitative Finance*, 8(3), 217-224.
- ▶ Gueant, O. (2016). *The Financial Mathematics of Market Liquidity*, Ch. 4. Chapman & Hall/CRC.
- ▶ Cartea, Jaikumar, & Penalva (2015). *Algorithmic and High-Frequency Trading*, Ch. 10.